MODULE 01 · EXAM STRATEGY - ALL DOMAINS

The ML and AI lifecycle

One arc. Every question sits somewhere on it. Learn to find the spot.

MODULE 01 of 43 DOMAIN All SCENARIOS 4

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Every scenario describes a system somewhere on **one arc**, from raw data to a decaying model.

Five stages, and the question each one answers

01
Frame
Is this even an ML problem, and which kind?

›

02
Prepare
Can the data support the model we want?

›

03
Develop
Does the model work well enough?

›

04
Deploy
Can it be called the way the business needs?

›

05
Operate
Is it still working, safely and affordably?

ON THE EXAM

Frame is not an exam domain, but it decides more answers than any single service does.

Is this even a machine learning problem?

- **ML needs a pattern in data** that is too complex or too changeable to write down.
- If the rule can be stated in a sentence, write the rule. A model is worse and costs more.
- If being wrong sometimes is unacceptable, ML is the wrong tool. Every model is wrong sometimes.
- **You need enough historical examples**, and they must resemble what you will see next.
- 'Assessing feasibility' is a named skill in the exam guide. It is tested as a real answer.

> ON THE EXAM
>
> A perfectly good exam answer is that the requirement does not need a model at all.

Naming the stage eliminates options before you compare features

- **Four plausible options** is the normal state of an exam question. That is the design.
- Options usually spread across stages. Only one stage matches the situation.
- Naming the stage kills two options **before** you compare any service features.
- Comparing features first is how you spend four minutes and still guess.
- It also protects you from the **most tempting distractor**: solving a nearby problem well.

> ON THE EXAM
>
> If two options sit at different stages, at least one of them is answering a question nobody asked.

How the stages map onto the four domains

STAGE	DOMAIN	WEIGHT
Frame	Spread across Domain 1 and Domain 2	-
Prepare	Domain 1 - Data Preparation for ML and AI	28%
Develop	Domain 2 - ML Model and FM Development	24%
Deploy	Domain 3 - Deployment and Orchestration	24%
Operate	Domain 4 - Operating, Monitoring, Securing	24%

ON THE EXAM

The arc is not a teaching device invented for this course. It is the syllabus, in order.

There are now **two paths** along that arc, and questions will mix them.

Traditional ML or foundation model?

THE TRADITIONAL PATH

You build the model

- You need labelled data, and a lot of it.
- Training is the expensive step.
- Accuracy is measured against known answers.
- The model does exactly one task.

THE FOUNDATION MODEL PATH

You adapt someone else's

- You need examples or documents, not labels.
- Inference is the expensive step.
- Quality is judged, not simply scored.
- One model, many tasks, via prompting.

THE TIEBREAKER

Where the money goes is the fastest tell. Traditional ML front-loads cost into training. Foundation models spread it across every single call.

The same idea, two vocabularies

TRADITIONAL ML	FOUNDATION MODEL EQUIVALENT
Feature engineering	Chunking, embedding, prompt design
Training on labelled data	Fine-tuning on prompt-response pairs
Retraining on fresh data	Refreshing the knowledge base
Accuracy, F1, RMSE	BLEU, ROUGE, BERTScore, LLM-as-a-judge
Data drift	Retrieval quality decay and prompt rot
Model registry and versions	Prompt versions and agent versions

ON THE EXAM

When a foundation model term is unfamiliar, ask what it is the equivalent of. The stage is usually the same.

SageMaker AI is not the same thing as SageMaker

- **Amazon SageMaker AI** is the service that builds, trains and deploys models.
- It was called Amazon SageMaker until **3 December 2024**. Same service, new name.
- **Amazon SageMaker** now means the wider platform for data, analytics and AI.
- That platform holds Unified Studio, Lakehouse, Catalog, **and** SageMaker AI.
- API namespaces were left unchanged, so your code and old docs still say sagemaker.

> ON THE EXAM
>
> The exam guide says SageMaker AI when it means training and endpoints. Read the name carefully.

Where SageMaker AI capabilities sit on the arc

STAGE	WHAT YOU WILL MEET
Prepare	Data Wrangler, Processing jobs, Feature Store, Ground Truth
Develop	Built-in algorithms, training jobs, JumpStart, automatic model tuning
Develop	Experiments, managed MLflow, Clarify, Model Registry
Deploy	Real-time, serverless, asynchronous endpoints, batch transform, Pipelines
Operate	Model Monitor, data capture, Inference Recommender, Savings Plans

ON THE EXAM

Four names here are in maintenance mode and closed to new customers. Still examinable. See module 38.

Amazon Bedrock, in one slide

- **Managed access to foundation models** through a single API. No infrastructure to run.
- Models from Amazon and several third parties. You choose per request.
- Customisation without training from scratch: **fine-tuning, continued pre-training, distillation**.
- **Knowledge Bases** for retrieval, **Guardrails** for safety, **Evaluations** for quality.
- Billed by **tokens**, not by instance-hours. That single fact changes every cost question.

> ON THE EXAM
>
> If a scenario says 'no ML expertise' or 'no infrastructure to manage' and involves language, Bedrock is usually in the right answer.

SageMaker AI or Amazon Bedrock?

REACH FOR SAGEMAKER AI WHEN

You control the model

- Prediction on your own tabular or images.
- You have labels and need a set accuracy.
- You need the artefact, or edge deployment.
- Steady traffic favours instance pricing.

REACH FOR BEDROCK WHEN

You consume a model

- Language, generation or reasoning tasks.
- You have documents, not labelled data.
- No one wants to manage infrastructure.
- Spiky traffic favours per-token pricing.

THE TIEBREAKER

Predicting a number or a class from your own data is SageMaker AI. Producing or understanding language is Bedrock.

Pre-trained AI services solve common problems outright

- **Textract** extracts text, forms and tables from documents. **Comprehend** reads it.
- **Transcribe** speech to text. **Polly** text to speech. **Translate** between languages.
- **Rekognition** for images and video. **Personalize** for recommendations.
- No training, no labels, no endpoints. An API call and a per-request charge.
- The catch: you get their model, not yours. Little control over edge cases.

> ON THE EXAM
>
> A question that names an everyday task and stresses low effort is usually pointing at one of these.

Climb this ladder from the bottom, not the top

01
AI service
Does Textract, Comprehend or Rekognition already do this?

›

02
Bedrock FM
Can a foundation model do it with a good prompt?

›

03
Customise
Add retrieval, or fine-tune on your examples.

›

04
JumpStart
Is there a pre-trained model to adapt?

›

05
Train
Only now: build and train your own.

ON THE EXAM

Each rung costs more effort than the one below it. Stop at the first rung that meets the requirement.

Choose the **most managed option** that meets the requirement, and no more.

Where the cost sits at each stage

- **Prepare.** Storage and processing. Usually the smallest line, and the easiest to ignore.
- **Develop.** GPU hours. Spiky, visible, and the one everybody watches.
- **Deploy.** The quiet killer. An idle endpoint bills all night and all weekend.
- **Operate.** Monitoring jobs, retraining runs, and on the FM path, tokens forever.
- Training cost is finite. Inference cost compounds for as long as the system lives.

> ON THE EXAM
>
> Most real ML bills are dominated by inference, not training. Cost questions on this exam reflect that.

The qualifier tells you which answer is wanted

IF THE QUESTION SAYS	IT IS ASKING FOR
LEAST operational overhead	The most managed service. Serverless beats managed beats self-run
MOST cost-effective	Cheapest that still meets every stated constraint
LEAST development effort	Least code written, even if it costs more to run
LEAST latency	Real-time paths only. Rules out anything batch
MOST operationally efficient	Fewest things for a human to do on an ordinary day

ON THE EXAM

In the bank analysed for this course, one question in four turned on a qualifier rather than on product knowledge.

What this exam will not ask you to do

OUT OF SCOPE, PER THE EXAM GUIDE	SO AN OPTION THAT REQUIRES IT IS SUSPECT
Design a full end-to-end architecture	Answers spanning six services to solve one stated problem
Set ML strategy or best practice for an org	Answers about governance frameworks nobody asked for
Work deeply in two or more ML domains	Answers needing NLP and computer vision expertise at once
Integrate a wide array of new tools	Answers built on third-party tooling over an AWS service

ON THE EXAM

Distractors are often built from real out-of-scope work. Recognising the boundary eliminates them quickly.

RAG or fine-tuning?

RETRIEVAL, WHEN

The problem is knowledge

- The model lacks facts it was never trained on.
- The content changes weekly, or daily.
- You need to cite where an answer came from.
- Access to documents differs per user.

FINE-TUNING, WHEN

The problem is behaviour

- You need a particular format, tone or style.
- You have many good example inputs and outputs.
- The task is narrow and stable over time.
- Prompts have grown long and are still not reliable.

THE TIEBREAKER

Knowledge problem, retrieve. Behaviour problem, fine-tune. Fine-tuning on facts that change is the classic wrong answer.

What an agent adds, and what it costs you

- An agent **chooses its own steps**. A pipeline follows steps you wrote.
- It calls tools, keeps state across turns, and decides when it is finished.
- New failure modes: tool call failures, coordination failures, truncated streaming.
- Cost becomes hard to predict, because the number of model calls is not fixed.
- Worth it when the path genuinely varies. Wasteful when the path is always the same.

> ON THE EXAM
>
> If you can draw the workflow as a flowchart, use Step Functions. If you genuinely cannot, that is when an agent earns its place.

Models decay. That is normal, not a bug

- **Data drift.** The inputs change. Yesterday's customers are not today's.
- **Concept drift.** The relationship changes. Fraud adapts to your detector.
- Neither shows up as an error. Latency is fine. Logs are clean. Predictions are worse.
- This is why Domain 4 is a quarter of the exam and not a footnote.
- The retrieval equivalent: documents change, retrieval quality quietly falls.

> ON THE EXAM
>
> A scenario where accuracy fell but nothing was deployed is always an Operate-stage question.

Four questions, in this order, on every scenario

01

Which stage?

Prepare, develop, deploy or operate. Cuts the options.

02

Which path?

Traditional model, or foundation model. Sets the vocabulary.

03

What is fixed?

The constraint they planted. Latency, cost, skills, compliance.

04

What is wanted?

The qualifier. Least overhead, least effort, lowest cost.

ON THE EXAM

Roughly fifteen seconds of reading, and it usually leaves two options standing instead of four.

> A retailer has run a demand-forecasting model on a SageMaker AI real-time endpoint for eight months. It was accurate at launch.
>
> Over the last quarter, forecasts have drifted steadily worse. No code has been deployed and no configuration has changed. The endpoint reports normal latency, no errors, and healthy CloudWatch metrics.

What should the team investigate first?

A Retrain the model immediately on the last twelve months of data

B Compare the recent inference inputs against the training data distribution

C Increase the endpoint instance size to reduce inference latency

D Roll the endpoint back to the previously deployed model version

◆ PAUSE AND DECIDE

What should the team investigate first?

A. Retrain the model immediately on the last twelve months of data

Might work, but it is a fix applied before the cause is known. If inputs shifted, retraining on unexamined data can bake the shift in.

B. Compare the recent inference inputs against the training data distribution

Diagnoses before treating. This is exactly the data-drift check, and it tells you whether retraining will even help.

C. Increase the endpoint instance size to reduce inference latency

Solves a problem the scenario explicitly ruled out. Latency was stated as normal.

D. Roll the endpoint back to the previously deployed model version

There is nothing to roll back to. No deployment happened. The model that degraded is the model that launched.

THE SIGNAL

Accuracy fell, nothing was deployed, infrastructure is healthy. That combination is always Operate stage, and always drift until proven otherwise.

> An insurance firm receives about four thousand scanned claim forms each day. They need the printed fields and the table of itemised charges pulled out and written to a database.
>
> The team has no machine learning engineers and no labelled training data. They want this running within a few weeks with the LEAST development effort.

Which approach should they take?

A. Use Amazon Textract to extract the fields and tables, then write the output to the database

B. Label a sample with SageMaker Ground Truth, then train an object detection model

C. Fine-tune a vision foundation model on Amazon Bedrock using the scanned forms

D. Train a custom convolutional neural network in SageMaker AI using script mode

◆ PAUSE AND DECIDE

Which approach should they take?

A Use Amazon Textract to extract the fields and tables, then write the output to the database

The bottom rung of the ladder. Textract extracts forms and tables from scanned documents with no training and no labels.

B Label a sample with SageMaker Ground Truth, then train an object detection model

Would eventually work, but it is weeks of labelling for a solved problem. It also fails the no-ML-engineers constraint.

C Fine-tune a vision foundation model on Amazon Bedrock using the scanned forms

The impressive answer, and wrong. Fine-tuning needs examples, expertise and evaluation, for a task a managed service already does.

D Train a custom convolutional neural network in SageMaker AI using script mode

The top rung of the ladder for a problem solved at the bottom rung. Maximum effort, and no team to do it.

THE SIGNAL

Three constraints were stated: no ML engineers, no labels, least development effort. Each one independently eliminates B, C and D.

> A company wants an internal assistant that answers staff questions about its HR policies. The policy documents are revised most weeks by the HR team, who are not engineers.
>
> Answers must cite the specific policy document they came from, so staff can check them.

Which approach best fits these requirements?

A. Fine-tune a foundation model on the policy documents, and repeat after each revision

B. Run continued pre-training on the policy corpus to embed the policies in the model

C. Paste the full policy set into every prompt as context

D. Build a retrieval system over the policy documents and ground the model's answers in it

◆ PAUSE AND DECIDE

Which approach best fits these requirements?

A. Fine-tune a foundation model on the policy documents, and repeat after each revision

A knowledge problem treated as a behaviour problem. Weekly revisions would mean weekly fine-tuning jobs, and it still cannot cite sources.

B. Run continued pre-training on the policy corpus to embed the policies in the model

Even heavier than fine-tuning, with the same two failures: stale on revision, and no citation.

C. Paste the full policy set into every prompt as context

Tempting because it is simple and always current. It breaks on context limits, and you pay for the entire policy set on every question.

D. Build a retrieval system over the policy documents and ground the model's answers in it

Content changes often, citation is required, and HR can update documents without an engineer. All three point the same way.

THE SIGNAL

Citation is the giveaway. Only retrieval can point at the passage an answer came from, because only retrieval fetched a passage.

> A bank scores its entire customer base for credit risk once a night. The job processes about two million records and must finish before the branches open in the morning.
>
> A junior engineer has proposed a SageMaker AI real-time endpoint with auto scaling, invoked in a loop by a Lambda function. The team asks you to review the design for cost.

What should you recommend?

A. Keep the real-time endpoint but add a scheduled scaling policy for the overnight window

B. Replace the design with an asynchronous inference endpoint

C. Replace the design with a SageMaker AI batch transform job

D. Keep the real-time endpoint but switch it to serverless inference

◆ PAUSE AND DECIDE

What should you recommend?

A. Keep the real-time endpoint but add a scheduled scaling policy for the overnight window

Improves a design that should not exist. You still pay for a persistent endpoint and still invoke two million times one at a time.

B. Replace the design with an asynchronous inference endpoint

Closer, and it does queue requests. But it is built for large individual payloads needing a response, not for scoring a whole dataset at once.

C. Replace the design with a SageMaker AI batch transform job

Nobody is waiting for an individual answer. Compute spins up, processes the dataset, writes to S3, and shuts down.

D. Keep the real-time endpoint but switch it to serverless inference

Removes idle cost but keeps the wrong shape: two million separate invocations, with cold starts, to produce one overnight file.

THE SIGNAL

Ask who is waiting. If the answer is nobody, real-time inference is the wrong shape however cleverly it is scaled.

What to take from this module

- **Name the stage first.** Prepare, develop, deploy, or operate. Then compare options.
- **Two paths, one arc.** Traditional models and foundation models, same five stages.
- **Climb the ladder from the bottom.** Most managed option that meets the requirement.
- **Knowledge problem, retrieve. Behaviour problem, fine-tune.**
- **Ask who is waiting.** If nobody is, real-time inference is the wrong shape.

> ON THE EXAM
>
> Module 02 starts Domain 1, where the data lives and how it reaches a training job.